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# Index

## 0.1 Exercise

### Exercise 0.1.1 ► Calculate

Let  $V$  be a vector space of finite dimension with basis  $(e_1, \dots, e_n)$  and let  $V^*$  denotes the dual space with dual basis  $(e_1^*, \dots, e_n^*)$ . Please prove in two different ways that for every  $(\alpha, \beta) \in \Lambda^p V^* \times \Lambda^q V^*$ ,

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

One way is to use definition, the other is to calculate in basis using the next exercise.

*Proof.* 1.

$$\alpha \wedge \beta = \frac{(p+q)!}{p!q!} \text{Alt}(\alpha \otimes \beta)$$

$$\beta \wedge \alpha = \frac{(p+q)!}{p!q!} \text{Alt}(\beta \otimes \alpha)$$

Only need to show that

$$\text{Alt}(\alpha \otimes \beta) = (-1)^{pq} (\beta \otimes \alpha)$$

Where

$$\text{Alt}(\alpha \otimes \beta) = \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) \sigma(\alpha \otimes \beta)$$

Let  $\tau = (p + 1, \dots, p + q, 1, 2, \dots, p)$ , then  $\text{sgn}(\tau) = (-1)^{pq}$ . We have

$$\begin{aligned}
 \text{Alt}(\alpha \otimes \beta) &= \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) \sigma \circ \tau (\beta \otimes \alpha) \\
 &= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) (\text{sgn} \tau) (\sigma \circ \tau) (\beta \otimes \alpha) \\
 &= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma' \in S_{p+q}} (\text{sgn} \sigma') \sigma' (\beta \otimes \alpha) \\
 &= (-1)^{pq} \text{Alt}(\beta \otimes \alpha)
 \end{aligned}$$

2. Suppose

$$\begin{aligned}
 \alpha &= \sum_{i_1 \dots i_p} a_{i_1 \dots i_p} e_{i_1}^* \wedge \dots \wedge e_{i_p}^* \\
 \beta &= \sum_{j_1 \dots j_q} b_{j_1 \dots j_q} e_{j_1}^* \wedge \dots \wedge e_{j_q}^*
 \end{aligned}$$

Then

$$\begin{aligned}
 \alpha \wedge \beta &= \sum_{i_1 \dots i_p} a_{i_1 \dots i_p} b_{j_1 \dots j_q} (e^*)_I \wedge (e^*)_J = \sum_{I,J} a_I b_J e_{IJ} \\
 \beta \wedge \alpha &= \sum_{I,J} a_I b_J e_{JI} = (-1)^{pq} \sum_{I,J} a_I b_J e_{IJ} = (-1)^{pq} \alpha \wedge \beta
 \end{aligned}$$

□

### Exercise 0.1.2

Let  $V$  be a vector space of finite dimension and let  $V^*$  denotes the dual space. Prove that if  $(\theta_1, \dots, \theta_r)$  is a family of  $r$  covectors in  $V^*$  which are **linearly dependent**, then the following wedge product:

$$\theta_1 \wedge \dots \wedge \theta_r = 0.$$

*Proof.* Since  $\theta_1 \wedge \dots \wedge \theta_r$  is multi-linear and alternative, take any vector  $v^1, \dots, v^r$ , we have

$$\theta_1 \wedge \dots \wedge \theta_r (v^1, \dots, v^r) = \det(\theta_i(v^j))_{ij}$$

Since  $\theta_1, \dots, \theta_r$  are linealy dependent , the row vectors of the matrix  $(\theta_i(v^j))_{ij}$  are as well. Thus  $\det(\theta_i(v^j))_{ij} = 0$  , and  $\theta_1 \wedge \dots \wedge \theta_r(v^1, \dots, v^r) = 0$ . Since  $v^1, \dots, v^r$  are arbitrary ,  $\theta_1 \wedge \dots \wedge \theta_r = 0$ .  $\square$

### Exercise 0.1.3 ► Basis of Grassmann algebra

Let  $V$  be a vector space of finite dimension with basis  $(e_1, \dots, e_n)$  and let  $V^*$  denotes the dual space with dual basis  $(e_1^*, \dots, e_n^*)$ . Please prove that

$$(e_{i_1}^* \wedge \dots \wedge e_{i_k}^*)_{1 \leq i_1 < \dots < i_k \leq n}$$

forms a basis of  $\Lambda^k V^*$ .

*Proof.* For  $\alpha \in \Lambda^k V^*$ , and  $I = (i_1, \dots, i_k)$ . Define

$$\alpha_I := \alpha(e_{i_1}, \dots, e_{i_k})$$

If  $I$  has a reputed index , then  $\alpha_I = 0$  , and if  $J = \sigma I$  ,  $\alpha_J = (\text{sgn} \sigma) \alpha_I$  . We have

$$\sum_I' \alpha_I e_I^*(e_{j_1}, \dots, e_{j_k}) = \sum_I' \alpha_I \delta_J^I = \alpha_J = \alpha(e_{j_1}, \dots, e_{j_k})$$

Thus  $\alpha = \sum_I' \alpha_I e_I^*$  . Which shows that  $e_{i_1}^*, \dots, e_{i_k}^*, 1 \leq i_1 < \dots < i_k \leq n$  spans the  $\Lambda^k V^*$ . To show that they are linealy independent. Suppose that

$$\sum_I k_I e_I^* = 0$$

Both sides act on  $e_J$ , we have  $k_J = 0$ .  $\square$

### Exercise 0.1.4 ► Grassmann algebra versus Polynomial functions on $V$ .

Let  $V$  be a vector space of finite dimension and let  $V^*$  denotes the dual space. A p-multilinear map  $\alpha$  on  $V$  is **symmetric** if

$$\alpha(v_1, \dots, v_n) = \alpha(v_{\sigma(1)}, \dots, v_{\sigma(n)})$$

for all permutations  $\sigma$ . Such  $\alpha$  defines a homogeneous polynomial  $\tilde{\alpha}$

defined by

$$\tilde{\alpha}(v) = \alpha(v, \dots, v)$$

for all  $v \in V$ . Then show that for symmetric multilinear maps  $\alpha, \beta$  on  $V$  of respective  $p, q$ , the product  $\alpha \circ \beta$  defined by

$$\alpha \circ \beta(v_1, \dots, v_{p+q}) := \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

is valued into symmetric multilinear map on  $V$  of degree  $p + q$  and that

$$\alpha \circ \beta(v, \dots, v) = \alpha(v, \dots, v) \beta(v, \dots, v).$$

Could you compare with  $\Lambda^\bullet V^*, \wedge$  ?

### Exercise 0.1.5 ► Basic important examples of smooth manifolds

Please prove that the following topological spaces have a smooth manifold structure :

- The Euclidean sphere  $S^n \subset \mathbb{R}^{n+1}$ .
- The matrix groups  $GL_n(\mathbb{R}), SL_n(\mathbb{R}), O_n(\mathbb{R})$ , these are Lie groups of matrices. Each time could you tell me what is the dimension of these groups.

*Proof.* • Define  $F$ :

$$F(x^1, \dots, x^{n+1}) = x_1^2 + \dots + x_{n+1}^2$$

Then

$$S^n = F^{-1}(1)$$

$$dF_x = \nabla F(x) = (2x_1, \dots, 2x_{n+1}) \neq 0, \forall x \in \mathbb{R}^{n+1}$$

By regular value theorem,  $S^n \subseteq \mathbb{R}^{n+1}$  is a regular embedded submanifold of  $\mathbb{R}^{n+1}$ , which itself is a manifold.

- 1.  $\text{Mat}(n, \mathbb{R}) \simeq \mathbb{R}^{n^2}$  是  $n^2$  维流形.
- 2.  $GL(n, \mathbb{R}) = \{M \in \text{Mat}(n, \mathbb{R}) : \det M \neq 0\}$  作为  $\text{Mat}(n, \mathbb{R})$  的一个开子集也是一个  $n^2$  维流形.

3.  $SL(n, \mathbb{R}) = \{M \in \text{Mat}(n, \mathbb{R}) : \det M = 1\}$

$$\begin{aligned}\det : \text{Mat}(n, \mathbb{R}) &\rightarrow \mathbb{R} \\ M &\mapsto \det M\end{aligned}$$

$SL(n, \mathbb{R}) = \det^{-1}(1)$ . 取一个  $M \in \text{Mat}(n, \mathbb{R})$ ,  $\det M = 1$ . 要证明  $(d(\det))_M : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$ . 按分量求偏导, 得到

$$\frac{\partial}{\partial a_{ij}} (\det M) = A_{ij} (\text{代数余子式})$$

由于  $\det M = 1$ . 至少有一个  $A_{ij} \neq 0$ . 从而  $(d(\det))_M$  是满的.

4. 定义  $P : \text{Mat}(n, \mathbb{R}) \rightarrow \text{Sym}(n, \mathbb{R})$  (对称矩阵)  $\simeq \mathbb{R}^{\frac{n(n+1)}{2}}$ .

$$P(A) := A^T A$$

则  $O(n, \mathbb{R}) = P^{-1}(0)$ . 对于  $A \in O(n, \mathbb{R})$ .

$$(dP)_A : \text{Mat}(n, \mathbb{R}) \rightarrow \text{Sym}(n, \mathbb{R})$$

与上例不同, 这里用分量的形式去计算 Jacobi 是看不清楚的, 不过由于  $P$  本身足够简单, 我们可以用微分最原始的含义来观察. 考虑

$$P(A + \varepsilon B) - P(A) = \varepsilon (dP)_A(B) + O(\dots)$$

带入得到

$$(A^T + \varepsilon B^T)(A + \varepsilon B) - A^T A = \varepsilon (B^T A + A^T B) + O(\varepsilon)$$

于是  $(dP)_A(B) = B^T A + A^T B$ . 只需要证明这个关于  $B$  的映射是满的. 为此, 解方程

$$A^T B + B^T A = C$$

取  $B = \frac{1}{2}AC$ , 利用  $A, C$  对称, 带入得到

$$\frac{1}{2}A^T AC + \frac{1}{2}C^T A^T A = \frac{1}{2}(C + C^T) = C$$

故  $O(n, \mathbb{R})$  是  $\text{Mat}(n, \mathbb{R})$  的一个  $\frac{n(n-1)}{2}$  维正则子流形. 注意到对于  $A \in O(n, \mathbb{R})$ ,  $\det A = 1$  或  $\det A = -1$ , 可以窥见事实上  $O(n, \mathbb{R})$  有两个连通分支.

□

### Exercise 0.1.6 ► Quotient manifolds

Prove that the following objects are quotient manifolds

- $\mathbb{R}P^n$  obtained by quotienting  $S^n$  by antipodal map, show that it is nonorientable by a picture. Could you explain by a picture that the neighborhood of the line at infinity is homeomorphic to a Möbius ribbon ?
- Prove that the torus  $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$  is a smooth manifold.

*Proof.*

- 1. See  $\mathbb{R}P^n$  is  $S^2/(x \sim -x)$ . The orientation changes at the equator. The map  $A : S^2 \rightarrow S^2, A(x) = -x$  induces an identity map on  $\mathbb{R}P^2$ ,  $dA$  send an orientation basis  $\{e_1, e_2\}$  to one opposite to it .
- 2. Use  $\mathbb{R}P^2 = D^2/\{x \sim -x \text{ for } x \in \partial D^2 = S^1\}$ . Then the neighborhood of the line at infinity is a thin ring near  $\partial D^2$  with some quotient relation. The quotient is just the same as that of Möbius .

□

### Exercise 0.1.7 ► Vector fields are derivation

Let  $M$  be a  $C^\infty$  manifold. Show that the left  $C^\infty$  module of derivation of the ring  $C^\infty(M)$  i.e. linear maps  $D : C^\infty(M) \rightarrow C^\infty(M)$  satisfying

$$D(fg) = (Df)g + f(Dg)$$

can be identified with vector fields on  $M$ , smooth sections of  $TM$ .

*Proof.* Define

$$D(x)f = (Df)(x)$$

Then

$$D(x)(fg) =$$

□

# Sard's Theorem

## 1.1 Set of Measure Zero

### Lemma 1.1.1

Suppose  $A \subseteq \mathbb{R}^n$  is a compact subset whose intersection with  $\{c\} \times \mathbb{R}^{n-1}$  has  $(n-1)$ -dimensional measure zero for every  $c \in \mathbb{R}$ . Then  $A$  has a  $n$ -dimensional measure zero.

*Proof sketch.*

把  $A$  的每一个“片”  $A_c := \{(c, x) \in \{c\} \times \mathbb{R}^{n-1} : (c, x) \in A\}$  的开覆盖, 缩短成里面的一个开的“方片”. 这些“方片”构成  $A$  的一个开覆盖, 取出一个有限子覆盖计算其 Volume, 将“方片”的相交部分抹去, 使得总体积可控.

*Proof.* Take  $[a, b] \subseteq \mathbb{R}$  such that  $A \subseteq [a, b] \times \mathbb{R}^{n-1}$ .

Let  $A_c := \{(c, x) \in \{c\} \times \mathbb{R}^{n-1} : (c, x) \in A\}$ . The projection from  $A$  to  $[a, b] \times \mathbb{R}^{n-1}$  is continuous map from a compact set to a Hausdorff space, thus is a proper map. It follows that  $A_c$  is a compact subspace of  $A$  for all  $c \in [a, b]$ .

Take any  $\delta > 0$ . There exists a finite open cover  $C_1, \dots, C_k$  of  $A_c$ , such that the total volume is less than  $\delta$ . Denote the open subset  $C_1 \cup \dots \cup C_k \subseteq A_c$  by  $U_c$ . We claim that there is an open interval  $J_c$  containing  $c$  and contained in  $U_c$ , such that  $A_c$  is contained in the intersection of  $A$  with  $J_c \times \mathbb{R}^{n-1}$ : otherwise, there is a sequence  $\{(c_i, x_i)\}$ , such that  $c_i \rightarrow c$ ,  $x_i \notin U_c$ ; but by passing to a subsequence  $x_i \rightarrow A_c \setminus U_c$ , which is a contradiction to  $A_c \subseteq U_c$ .

The intervals  $\{J_c : c \in [a, b]\}$  form an open cover of  $[a, b]$ , so there is a finite many  $c_1 < c_2 < \dots < c_m$  such that  $J_{c_1}, \dots, J_{c_m}$  forms a finite open cover of  $A$ . We may assume that the total length is less than  $2|b-a|$  by shrinking the intersection of the members. Then  $J_{c_1}, \dots, J_{c_m}$  is an open cover of  $A$  with total volume less than  $2\delta|b-a|$ , which can be arbitrarily small. Hence  $A$  has a  $n$ -dimensional measure zero.  $\square$



### Proposition 1.1.2

Suppose  $A$  is an open or closed subset of  $\mathbb{R}^{n-1}$  or  $\mathbb{H}^{n-1}$ , and  $f : A \rightarrow \mathbb{R}$  is a continuous function. Then the graph of  $f$  has measure zero in  $\mathbb{R}^n$ .

*Proof sketch.*

对于  $A$  是紧集的情况, 令  $P^n = \{\text{Sets are } n\text{-dimensional measure zero}\}$ . 上面的引理给出了  $\forall c, P_c^{n-1} \implies P^n$  的结论. 因此降维操作是无损的, 从而归纳法有效.

*Proof.* First assume that  $A$  is compact. We prove the theorem in this case by induction on  $n$ . When  $n = 1$ , it is trivial because the graph of  $f$  is at most a point. To prove the inductive step, we appeal to lemma above. For each  $c \in \mathbb{R}$ , the intersection of the graph of  $f$  with  $\{c\} \times \mathbb{R}^{n-1}$  is just the graph of the restriction of  $f$  to  $\{x \in A : x^1 = c\}$ , which is in turn the graph of continuous functions of  $(n-2)$  variables. It follows by induction that each such graph has  $(n-1)$ -dimensional measure zero, and thus by lemma above, the graph of  $f$  itself has  $n$ -dimensional measure zero.

If  $A$  is noncompact, it is a countable union of compact subsets, so the graph of  $f$  is a countable union of sets of measure zero.  $\square$

### Corollary 1.1.3

Every proper affine subspace of  $\mathbb{R}^n$  has measure zero in  $\mathbb{R}^n$ .

### Proposition 1.1.4 ▶ Diffeomorphism-invariant of Measure Zero

Suppose  $A \subseteq \mathbb{R}^n$  has measure zero and  $F : A \rightarrow \mathbb{R}^n$  is a smooth map. Then  $F(A)$  has measure zero.

### Definition 1.1.5

If  $M$  is a smooth  $n$ -manifold with or without boundary, we say that a subset  $A \subseteq M$  has **measure zero in  $M$**  if for every smooth chart  $(U, \varphi)$  for  $M$ , the subset  $\varphi(A \cap U) \subseteq \mathbb{R}^n$  has  $n$ -dimensional measure zero.

### Lemma 1.1.6

Let  $M$  be a smooth  $n$ -manifold with or without boundary and  $A \subseteq M$ . Suppose that for some collection  $\{(U_\alpha, \varphi_\alpha)\}$  of smooth charts whose

domains cover  $A$ ,  $\varphi_\alpha(A \cap U_\alpha)$  has measure zero in  $\mathbb{R}^n$  for each  $\alpha$ . Then  $A$  has measure zero in  $M$ .

*Proof.* Let  $(V, \psi)$  be any smooth chart for  $M$ . We need to show that  $\psi(A \cap V)$  has measure zero. There exists countably many  $U_\alpha$ 's covering  $A \cap V$ . We write

$$\psi(A \cap V \cap U_\alpha) = (\psi \circ \varphi_\alpha^{-1}) \circ \varphi_\alpha(A \cap V \cap U_\alpha)$$

$\varphi_\alpha(A \cap V \cap U_\alpha)$  is a subset of measure zero set  $\varphi_\alpha(A \cap U_\alpha)$ , thus has measure zero as well. It follows from Proposition ?? that  $\psi(A \cap V \cap U_\alpha)$  has measure zero. Then  $\psi(A \cap V)$  is a countable union of measure zero sets, thus is measure zero as well.  $\square$

### Exercise 1.1.7

Let  $M$  be a smooth manifold with or without boundary. Show that a countable union of sets of measure zero in  $M$  has measure zero.

*Proof.* Suppose  $A_1, A_2, \dots$  are sets of measure zero in  $M$ . Let  $A = \bigcup_{i=1}^{\infty} A_i$ . Take charts  $(U_{ij}, \varphi_{ij})$ ,  $j = 1, 2, \dots$  such that  $A_i \subseteq \bigcup_{j=1}^{\infty} U_{ij}$ . Then  $\{U_{ij}\}_{i,j}$  are countably many open sets that containing  $A$ , such that for each  $i, j$ ,

$$\varphi_{ij}(A \cap U_{ij}) \subseteq \bigcup_{i=1}^{\infty} \varphi_{ij}(A_i \cap U_{ij})$$

is a countable union of measure zero sets, thus itself is measure zero. Then it follows from Lemma ?? that  $A$  is of measure zero.  $\square$

### Proposition 1.1.8

Suppose  $M$  is a smooth manifold with or without boundary and  $A \subseteq M$  has measure zero in  $M$ . Then  $M \setminus A$  is dense in  $M$ .

*Proof.* If  $M \setminus A$  is not dense. Then  $A$  contains a nonempty open subset of  $M$ , which implies that there is a chart  $(V, \psi)$ , such that  $\psi(A \cap V)$  is a nonempty open subset of  $\mathbb{R}^n$ . It is impossible that  $\psi(A \cap V)$  has measure zero at the same time that it is open.  $\square$

# Integral Curves and Flows